

that make learning easier. Giriyaapur says, "We are further bringing in some more equipment that could help students develop their skills."

New era of labs

While simulation based labs are set up based on a mathematical base, there are also experiment based labs where experiments are conducted and students get to have a feel of the data. Third type is the remote lab, where an actual lab is made available to users through any device connected through the Internet, iPhone, Android or a PC.

Essentially a laboratory controlled over the Internet, remote labs are designed exclusively for learning and monitoring any experiment integrated into the college lab. Electrono Solutions is one firm providing this solution in India.

On the virtual lab side is the *desi* Sakshat Virtual Labs started by Ministry of Human Resource and Development (MHRD). The team states on the website that the objective of this lab is to perform virtual experiments in basic electronics and yet have a close-to-real-life experience. The tool is designed such that students will get a feel of the real lab as its equipment, components and behaviour is mimicked by the system.

The virtual lab set up at NIT Su-

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rathkal is aptly named SOLVE—an acronym for Students Online Lab for Virtual Experimentation. NIT Surathkal alone has 70,000 users from all over the world at its virtual website (solve.nitk.ac.in), including a few thousand from Pakistan.

Various areas covered by these virtual labs include field-programmable gate arrays, basic electronics and others. For instance, IIT Delhi has a virtual wireless lab that gives students an actual experiment to learn from and keep modifying the antennae available to check propagation testing.

M-learning solutions

M-learning solutions ranging from Edula for basic aptitude learning to Eduvance for electronics hardware development, among others, are now helping students to efficiently utilise the time spent on their phones by

increasing their skills. These solutions consist of a simple application that can be installed on a smartphone, and learning happens on the go.

Is there such a thing as too many tools

"Engineering is supposed to solve problems," says Brijesh. A person who considers a challenge can smartly develop problem-solving skills. While it is granted that smart solutions make learning easier, is it sufficient to develop the required core skills for an engineer?

While some colleges are just getting on the bandwagon of bringing in development boards, others are trying to wean students off the development boards once the initial project is done, so that they develop more core engineering skills by building their own solutions from scratch. Iyer adds that tools like development boards tend to attract students to the wonders of engineering on one side, while on the other point them towards developing proficiency in software rather than base hardware.

A proper mix and match of both seems to be the best formula. Students are first taught how to rapidly prototype working projects using tools, and as the next step learn how to build projects from scratch.

Dr Joshi recalls his student days with, "We used to have a designated class just to learn how to work with lab equipment." Such additions in schedule of students can also help overcome the calculator-oscilloscope jump. **EFY**

Getting the syllabus up to speed

Many firms now have dedicated programmes such as Cadence' VLSI certification programme that aims to bridge the knowledge gap that takes a person from being a graduate to industry-ready. LDRA runs their own LDRA Academic Alliance Program, which aims to educate scholars generate a competitive and experienced workforce by helping in lab setups, syllabus updated and faculty-development programmes.

Pathy Iyer, head of business development at Keysight Technologies, says, "There is a government push to bridge the gap between academia and industry. The government feels that instead of completely changing the syllabus, we first do a short-term course to get them ready." Keysight with their knowledge of the wireless ecosystem has already started working with colleges.

NITs have also been asked to revamp the syllabus with the aim to bridge the industry-academia gap. "We are already talking with a few NITs to take the lead," adds Iyer.

Bharat Electronics, too, is focused on student collaboration. Dr Ajit Kalghatgi, chief scientist, CRL, Bharat Electronics, says, "We have student tie-ups with IISC and a few IIT projects where we define the area of work and jointly put some teams along with IISC or IIT professors, and at the end of it there is some technology. It could even be a published paper. Then it takes some more effort by the industry to convert that research into a product. This has been ongoing for many years; in our R&D effort about 87 papers were published."